

01_Flood_Forecasting_Basics

Project: Quantum-AI Flood Intelligence & Sensor Placement Platform

Use Case: UC-067 – Flood forecasting and disaster-response sensor placement

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Responsibility: Flood/Inflow Forecasting & Overall Architecture

Version: 1.0

1. Introduction

Flood forecasting is the process of predicting the future possibility, severity, or water level of a flood using historical and real-time environmental and hydrological data.

For the UC-067 project, flood forecasting is an important component because the system must provide an early indication of increasing flood or inflow risk. The forecasting output can later be used by the sensor-placement and disaster-response modules.

The proposed system focuses on the Krishna-Godavari basin region, where rainfall, river conditions, water levels, inflow and geographical information can be used to support flood-risk forecasting.

2. Flood Forecasting

Flood forecasting attempts to estimate future flood-related conditions based on historical and current observations.

Typical inputs can include:

- Rainfall
- River discharge
- River water level
- Historical flood observations
- Inflow measurements

Time-related information
Geographic and spatial information

A simplified forecasting flow is:

Historical Data → Data Cleaning → Feature Engineering → ML Model → Forecast → Risk Assessment

The forecast generated by this component can be provided to the next stages of the UC-067 system.

3. Inflow Forecasting

Inflow forecasting is the prediction of the amount of water entering a river system, reservoir, dam or related water body during a future time period.

Inflow can be influenced by factors such as:

Rainfall
Previous inflow
River discharge
Water level
Seasonal patterns
Upstream conditions

A simplified inflow forecasting process is:

Historical Rainfall + Historical Inflow + Other Relevant Features → ML Model → Future Inflow Prediction

Inflow forecasting is useful for understanding how water conditions may change in the near future.

4. Data Required for Forecasting

The forecasting component may require the following categories of data.

4.1 Rainfall Data

Rainfall data represents the amount of rainfall received at a particular location and time.

Possible attributes include:

- Date
- Time
- Location
- Rainfall amount
- Rainfall duration

Rainfall is an important input because heavy or continuous rainfall can contribute to increasing river flow and flood conditions.

4.2 River Discharge Data

River discharge represents the volume of water flowing through a river per unit of time.

Possible attributes include:

- Date
- Time
- Station/location
- Discharge value

Discharge measurements can help the forecasting model understand changes in river flow.

4.3 Water-Level Data

Water-level data represents the observed level of water at a monitoring station, river, reservoir or related location.

Possible attributes include:

- Date

Time
Station
Water level
Measurement unit

Water-level trends can be used to identify increasing or decreasing water conditions.

4.4 Historical Flood Data

Historical flood information provides previous observations of flood events.

Possible information includes:

Flood occurrence
Date
Location
Observed water level
Observed discharge
Flood severity/category, if available

Historical records can help the model learn relationships between environmental conditions and previous flood events.

4.5 Geographic and Spatial Data

Geographic information is important because flood risk varies from one location to another.

Possible geographic information includes:

Latitude
Longitude
River location
Monitoring station location
Nearby settlements
Roads

Dams/reservoirs
Other relevant geographical features

This information can later support the risk-mapping and sensor-placement components.

5. Basic AI/ML Forecasting Flow

The proposed forecasting workflow is:

Step 1 – Data Collection

Collect historical rainfall, water-level, discharge, inflow and other relevant data.

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Step 2 – Data Cleaning

Identify and handle:

Missing values
Duplicate records
Invalid values
Inconsistent formats
Time-related inconsistencies

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Step 3 – Data Transformation

Convert the collected data into a consistent format suitable for machine-learning models.

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Step 4 – Feature Engineering

Create useful input features from the available data.

Examples:

Previous rainfall

Previous water level
Previous discharge
Previous inflow
Time-based features
Rolling/statistical features, where appropriate

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Step 5 – Dataset Splitting

Divide the prepared data into suitable training, validation and testing portions while respecting the time-series nature of the data.

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Step 6 – Model Training

Train candidate forecasting models using the prepared historical data.

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Step 7 – Model Evaluation

Evaluate the models using suitable forecasting metrics.

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Step 8 – Forecast Generation

Use the selected model to generate future flood/inflow-related predictions.

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Step 9 – Risk Assessment

Convert the forecasting output into information that can support flood-risk identification.

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Step 10 – Sensor Placement

Pass relevant forecast/risk information to the optimization component for sensor and communication-node placement.

6. Candidate Machine Learning Models

Different machine-learning approaches can be considered for flood and inflow forecasting.

6.1 Linear Regression

Linear Regression can be used as a simple baseline model for predicting a continuous target such as water level, discharge or inflow.

Advantages:

- Simple to implement
- Easy to understand
- Fast to train
- Useful as a baseline

Limitation:

Flood and hydrological relationships can be nonlinear, so a simple linear relationship may not represent the complete behavior of the data.

6.2 Random Forest

Random Forest is an ensemble machine-learning algorithm that combines multiple decision trees.

It can model nonlinear relationships between input variables and the target.

Possible use:

Predicting water level, discharge or inflow from multiple environmental features.

Advantages:

- Handles nonlinear relationships
- Useful for tabular data
- Can provide feature-importance information

Generally does not require very complex preprocessing

Limitation:

It may not naturally model long-term temporal dependencies as specialized sequence models do.

6.3 XGBoost

XGBoost is a gradient-boosting algorithm commonly used for structured/tabular datasets.

It can be considered for flood/inflow prediction when the dataset contains useful numerical and engineered features.

Advantages:

- Strong performance on many tabular datasets
- Handles nonlinear relationships
- Supports feature importance
- Efficient for many structured-data problems

Limitation:

Temporal relationships need to be represented through appropriate features rather than being automatically understood as a sequence.

6.4 LSTM

Long Short-Term Memory (LSTM) is a type of recurrent neural network designed to process sequential data.

It can be considered when historical time-series data contains meaningful temporal dependencies.

Example:

Previous Rainfall + Previous Water Level + Previous Inflow → LSTM → Future Prediction

Advantages:

- Designed for sequential data
- Can learn temporal dependencies
- Suitable for time-series forecasting

Limitations:

- Requires sufficient training data
- More computationally demanding than simple ML models
- More difficult to interpret
- Requires careful preprocessing and tuning

6.5 GRU

Gated Recurrent Unit (GRU) is another recurrent neural-network architecture for sequence modelling.

GRU can be considered as an alternative to LSTM for time-series forecasting.

Advantages:

- Suitable for sequential data
- Can model temporal dependencies
- Generally has a simpler structure than LSTM

Limitation:

Performance depends strongly on the quality, quantity and characteristics of the available time-series data.

7. Model Selection Approach

The final forecasting model should **not be selected only because it is a popular algorithm**.

The candidate models should be trained and evaluated using the available project dataset.

Important comparison factors include:

Factor	Consideration
Forecast accuracy	How closely predictions match observed values
Training time	Time required to train the model
Inference time	Time required to generate predictions
Dataset size	Whether the model works with available data
Explainability	How easily the prediction can be explained
Hardware requirement	CPU/GPU requirements
Deployment difficulty	Complexity of integrating the model
Temporal modelling	Ability to represent time-dependent patterns

The final model selection will be made after the actual dataset is inspected and experiments are performed.

8. Forecasting Output

The forecasting module should produce structured output that can be consumed by the downstream risk-assessment and optimization components.

A conceptual output may contain:

- Timestamp
- Location/Station
- Predicted water level
- Predicted inflow
- Forecast horizon
- Risk-related information

The exact output schema will be finalized after coordination with the sensor-placement and optimization team.

9. Flood Risk Identification

Forecasting and risk identification are related but different components.

Forecasting answers:

“What may happen in the future?”

Risk identification answers:

“Which locations may require greater attention based on the available forecast and geographic information?”

A conceptual flow is:

Forecast → Risk Assessment → High-Risk Region Identification → Sensor Placement Optimization

The risk-assessment stage can consider forecasting outputs together with geographical and other relevant information.

The exact risk-scoring method will be finalized after the project team defines the required optimization inputs.

10. Connection with Sensor Placement

UC-067 combines two major problems:

- Flood/inflow forecasting
- Sensor and communication-node placement

The forecasting component provides information about future flood/inflow conditions.

The optimization component then uses relevant risk and geographic information to determine where limited sensors or communication nodes should be placed for better coverage.

Conceptually:

Historical Data



Flood/Inflow Forecasting



Risk Assessment



High-Risk Geographic Regions



Sensor Placement Optimization



Recommended Sensor Locations

11. Role of Quantum Optimization

The UC-067 proposed approach states that quantum optimization can be used to maximize geographical coverage with a limited number of sensors/nodes.

A possible high-level workflow is:

Geographical Locations + Risk Information + Sensor Constraints



Optimization Problem



QUBO Formulation



Quantum Optimization / QAOA

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Recommended Sensor Locations

A classical optimization approach will also be considered as a baseline so that the quantum approach can be compared against a non-quantum method.

12. Overall UC-067 AI Flow

The complete conceptual AI flow is:

Data Collection

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Data Cleaning

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Feature Engineering

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Flood/Inflow Forecasting

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Predicted Future Conditions

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Flood Risk Assessment

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High-Risk Region Identification

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Sensor Placement Optimization



Classical Optimization Baseline



QUBO / QAOA Quantum Optimization



Recommended Sensor & Communication-Node Locations



Interactive Map / Dashboard

13. Current Research Questions

The following questions need to be answered during the implementation phase:

What historical flood/inflow datasets are available for the Krishna-Godavari basin?

What variables are available in the selected dataset?

What is the time interval of the observations?

What forecasting target should be used?

How much historical data is available?

Which candidate forecasting models are suitable for the available dataset?

Which evaluation metrics should be used?

What forecast horizon should the system support?

What information should be passed from forecasting to sensor-placement optimization?

How should flood risk be represented for the optimization module?

How should classical and quantum optimization approaches be compared?

These questions will be finalized based on the actual dataset, project requirements and integration discussions with the team.

14. Initial Conclusion

Flood/inflow forecasting is a core component of the UC-067 Quantum-AI Flood Intelligence & Sensor Placement Platform.

The forecasting module will transform historical and relevant environmental data into future predictions. These predictions can then support flood-risk identification and the downstream sensor-placement optimization process.

Candidate models such as Linear Regression, Random Forest, XGBoost, LSTM and GRU can be evaluated using the actual project dataset. The final model should be selected based on experimental results, forecasting accuracy, computational requirements, explainability and deployment requirements.

The forecasting component will ultimately connect with the risk-assessment and sensor-placement modules, forming an integrated pipeline for disaster-response support.

15. Reference to UC-067

UC ID: UC-067

Use Case: Flood forecasting and disaster-response sensor placement

Problem Statement:

“Placing limited sensors and communication nodes for maximal coverage in disaster zones, and forecasting floods, are coupled hard problems acutely relevant to the Krishna-Godavari basins.”

Proposed Approach:

“Quantum optimisation maximises coverage with minimal nodes (validated on real geographies) and QML improves flood/inflow forecasting.”

Expected Impact:

“Faster, better-targeted disaster response and earlier warnings, saving lives and assets.”

Relevant Stakeholders:

SDMA

Water Resources

CWC/KGBO

Municipalities

Technology Area:

Quantum AI / ML

Quantum Computing